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A closed-loop framework integrating robotic inspection and digital twins for fatigue prognosis of in-service steel bridges

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Xincheng Li[1,2], Zhongqiu Fu[1] ✉, Hongbin Guo[1], Bohai Ji[1] ✉ & Zhaodong Xu[3]

Fatigue cracking threatens the safety of orthotropic steel decks (OSDs) in long-span bridges, yet current manual inspection lacks predictive depth. We present a closed-loop framework integrating autonomous robotic inspection, vision-based quantification, and finite-element fracture mechanics to enable adaptive fatigue prognosis. In laboratory validation, the robotic platform achieved a mean localization accuracy of 2.7 ± 0.8 cm, meeting structural precision requirements. Field deployment on an in-service cable-stayed bridge demonstrated that automated inspection reduced average time per girder from 124.6 to 50.4 minutes—a 59.6% reduction. Identified cracks were assimilated into a digital twin for adaptive state updating. Analysis of discrepancies between simulated and observed propagation paths—interpreted via stress intensity factor fields—highlighted significant mixed-mode fracture effects, particularly elevated Mode-II (shear) contributions, as a primary source of predictive uncertainty under in-service conditions. This integration of robotics and digital twins provides a scalable solution for automated maintenance. Beyond labor reduction, the framework establishes a data-driven path toward proactive life-cycle management, enhancing the structural resilience and long-term safety of critical transportation infrastructure.

Orthotropic steel decks (OSDs) have become the preferred structural form for long-span steel bridges owing to their high strength-to-weight ratio, modular construction efficiency, and favorable structural performance[1]. However, welded joints in OSDs are highly susceptible to fatigue damage under repetitive traffic loading, posing a serious threat to structural integrity[2]. Fatigue cracking at critical connections, particularly deck-rib and diaphragm-rib welds[3,4], has emerged as one of the predominant failure modes in aging infrastructure. Field investigations indicate that steel bridges with over 10 years of service can develop more than 2000 fatigue cracks per structure. Such extensive damage directly undermines load-bearing capacity and has become a critical bottleneck for safe and reliable service[5].

Conventional structural health monitoring (SHM) systems focus on global bridge responses such as displacement and vibration[6]. While effective for detecting global anomalies, they are limited in identifying localized fatigue damage in OSDs, where the initiation sites are spatially unpredictable and temporally uncertain[7]. Fixed sensor networks for full-bridge coverage are often economically infeasible and still fail to ensure complete detection[8]. Consequently, engineers still rely heavily on manual inspection for fatigue evaluation. This process requires temporary scaffolding, visual crack identification by trained inspectors, and manual documentation. Within confined steel box girders, such procedures are inefficient, prone to subjective error, and limited in inspection frequency[9].

Recent advances in robotics have created new opportunities for automated bridge inspection. Various robotic platforms have been proposed, including bio-inspired climbing robots[10], unmanned aerial vehicles[11], ground vehicles[12], and magnetic adhesion climbing robots[13]. Among them, magnetic wheeled robots are particularly promising for navigating inside steel box girders due to their stability and access to weld details[14]. To enhance detection accuracy and reduce operator dependence, camera-based inspection has been increasingly adopted[15], applied to quantify crack geometry[16,17]. Robotic inspection has further benefited from the integration of high-resolution cameras and NDT devices[18,19], enabling systematic data collection. Despite these advances, existing studies primarily aim to improve crack recognition precision rather than integrating results into system-level

[1]College of Civil and Transportation Engineering, Hohai University, Nanjing, China. [2]Chemical and Materials Engineering, University of Alberta, Edmonton, AB, Canada. [3]School of Civil Engineering, Southeast University, Nanjing, China. ✉e-mail: fuzhongqiu@hhu.edu.cn; hhbhji@163.com

assessments[20]. As a result, identified cracks are often treated as isolated defects without establishing their spatial relation to the global structure. Consequently, performance assessment and maintenance decisions still depend on manual interpretation[21]. Automated detection of fatigue damage in OSDs is thus a critical prerequisite for dynamic structural state updating with minimal human intervention.

Finite element modeling (FEM) has long played a central role in evaluating the fatigue life of OSDs[22]. Advances in FEM have spurred growing interest in fatigue life prediction, underscoring its utility and accuracy[23]. Building upon these capabilities, digital twin (DT) frameworks have been developed to simulate and assess the fatigue performance of OSDs[24]. By integrating sensor data into advanced FEM and AI-driven models, DTs enable dynamic interaction between physical structures and their virtual counterparts[25]. These developments create new opportunities for crack detection, performance assessment, lifecycle management, and SHM[26]. However, inspection technologies and DT models are often developed independently, resulting in a disconnect between the demand for accurate DT reconstruction and the information provided by inspections. This gap necessitates manual intervention for DT updating[27]. Current DT models typically serve diagnostic or predictive roles[28], yet discrepancies remain between simulated and observed crack propagation under complex in-service conditions[29]. Without a feedback mechanism to reconcile these differences, DTs risk diverging from actual structural behavior, thereby limiting their value for proactive decision-making. Although DTs have been widely explored for bridge monitoring and fatigue prognosis[30], most existing frameworks are validated only on Mode-I-dominated cracks or laboratory-scale specimens. Their applicability to complex, mixed-mode fatigue cracks in in-service steel structures has not been quantitatively verified.

Fatigue damage in OSDs has been widely reported in field observations and experimental studies. Typical cracking locations include welded joints such as the deck-rib joint and diaphragm-rib joint[31], as well as other geometric discontinuities[32], where local stress concentrations are pronounced[33]. Crack initiation and propagation are commonly attributed to wheel-load-induced local bending and secondary stresses, weld joint stress concentration, geometric stiffness discontinuities, and interactions with welding-induced residual stresses and fabrication imperfections[34]. While extensive work has been conducted on fatigue hotspot identification[35], local detail modeling[36], and fatigue assessment of OSDs[37], practical in-service management is still challenged by limited inspection frequency and the difficulty of continuously updating crack geometry and loading conditions in a physics-based framework[38].

To address these challenges, we propose and validate a closed-loop fatigue evaluation system tailored for in-service steel bridges. The system integrates three synergistic layers: autonomous damage acquisition, dynamic DT updating, and decision-support interaction. Guided by fatigue hot-spot statistics, the robotic inspector performs coverage-based inspections of fatigue-critical details, quantifies cracks from the acquired images, and outputs pose-referenced crack skeletons in the global structural frame. These OBS cracks are embedded into high-fidelity FEM models, coupled with traffic load spectra to simulate crack propagation and predict remaining fatigue life. Maintenance decisions are triggered based on the predicted remaining life. Meanwhile, deviations between predicted and observed crack morphologies are quantitatively analyzed to inform the model, forming a closed adaptive loop that refines both fatigue assessment and monitoring strategies.

Methods

System architecture

To evaluate the fatigue performance of in-service OSDs, a three-layer DT framework was implemented, comprising: (1) damage acquisition through robotic inspection, (2) digital-twin updating via finite-element simulations, and (3) fatigue state assessment, as shown in Fig. 1. The following subsections present its quantitative validation from laboratory to field scale.

Damage acquisition (physical layer): an autonomous robotic platform equipped with a manipulator and high-resolution camera navigates inside the steel box girder along predefined paths. The robot acquires weld-line images, performs real-time crack segmentation, and registers three-dimensional crack skeletons $[x_i, y_i, z_i]$ through multi-sensor localization, thereby enabling automated, quantitative fatigue damage sensing.

DT updating (virtual layer): a high-fidelity finite element (FE) model of the OSDs is constructed from as-built geometry. Crack skeletons extracted from the physical layer are mapped into the model, followed by sub-model partitioning and mesh refinement around crack tips. Traffic loading spectra are applied, and fracture simulations are conducted to propagate cracks and evaluate structural response in the damaged state. The model architecture is open and extensible, allowing the integration of additional influencing factors such as residual stresses or material heterogeneity.

Fatigue state assessment (interaction layer): by combining real fatigue damage observations with DT crack predictions, the framework enables two levels of decision support: (i) maintenance decision-making, where DT crack evolution informs targeted repair or strengthening actions; and (ii) feedback correction, where deviations between predicted and observed crack paths are quantified in terms of stress intensity factor differences, guiding model calibration and the optimization of monitoring strategies.

This paper discusses three types of crack morphologies: (a) ground truth (GT): manually measured labels used only for algorithm validation. (b) observation baseline (OBS): geo-registered crack geometry obtained from robotic inspection. (c) digital-twin predicted (DT): crack geometry predicted DT-based model.

Fatigue-informed inspection strategy

A clear understanding of actual fatigue crack distributions in OSDs requires real-world data collection[39]. Given that more than 95% of fatigue cracks in OSDs occur at the deck-rib and diaphragm-rib welded joints[40], we designed the inspection strategy to target these fatigue-prone regions, as shown in Fig. 2a. Accordingly, the inspection paths of the robot covered both deck-rib weld lines and diaphragm-rib welds. Routes along adjacent U-ribs allowed full coverage of deck-rib welds, while diaphragm paths provided access to diaphragm-rib joints and also served as transition routes between ribs, as shown in Fig. 2b. Unlike conventional path planning that primarily minimizes travel distance or time, we first define the fatigue-prone welded details, and then design the inspection route under the robot's reachability constraints inside the closed box girder. The resulting route is therefore coverage-driven rather than globally kinematics-optimal, and may be further optimized in future work. This prioritization does not imply that other crack types cannot occur; rather, deck-rib and diaphragm-rib welds are targeted first due to their dominance in observed cracking, and the same inspection and localization pipeline can be extended to additional details by adding corresponding target regions.

To implement the above strategy, we developed a magnetic-wheeled robotic platform capable of autonomous movement and crack image acquisition within closed steel box girders, as shown in Fig. 2c. The system consists of five main modules: chassis, locomotion, localization, image acquisition, and control/processing. The chassis was fabricated from 2 mm-thick aluminum with front and rear guiding frames to facilitate wall-to-wall transitions. The locomotion system employed six independently driven NdFeB permanent-magnet wheels, providing stable adhesion and maneuverability on steel surfaces. Each wheel was powered by an individual motor, enabling forward, backward, and turning motions.

The robot dimensions were optimized to traverse both deck plates and diaphragms. The localization module combined an inertial measurement unit (IMU), lidar, and wheel encoders, while the imaging system integrated a three-degree-of-freedom robotic arm carrying a high-resolution RGB camera. Each joint of the robotic arm provided fine angular control, allowing the camera to orient toward weld lines at designated inspection positions. The control and computation system included an STM32 microcontroller for motion control and a Raspberry Pi for image processing and localization. Together, these modules enable autonomous execution of the inspection route and automatic geo-tagging of the collected images/videos using the estimated robot pose. Crack locations are not assumed to be

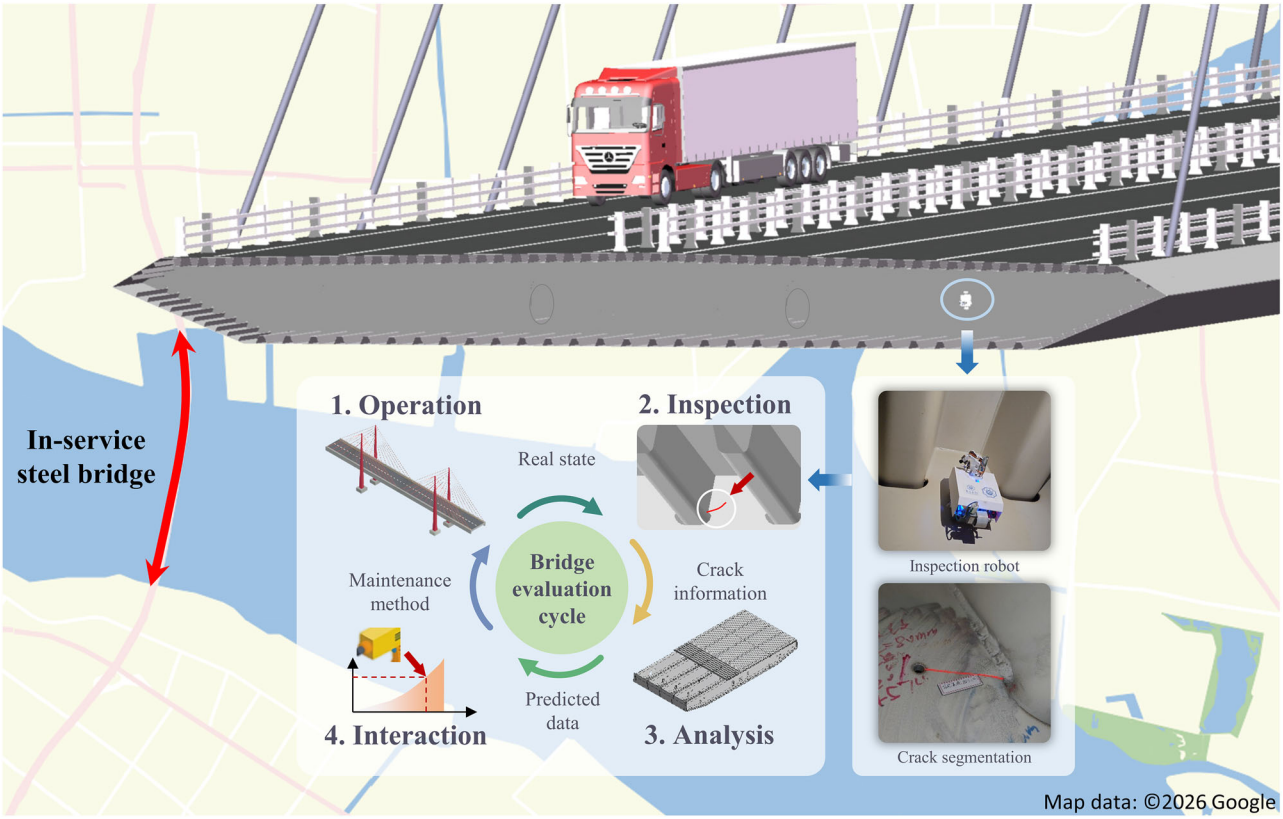


Fig. 1 | Scheme of automatic orthotropic steel decks (OSDs) fatigue performance update system for in-service bridge. The scheme involves collecting fatigue crack image data through automated inspection robot of in-service bridges, and analyzing crack propagation behavior using digital twin model, thereby providing support for bridge maintenance methods.

known prior to inspection. The robot performs a coverage-based scan of fatigue-critical details, while crack localization is completed after data acquisition. Specifically, crack skeleton nodes extracted from images are mapped into the component coordinate system using the recorded robot pose and arm calibration, yielding the spatial crack location (see Sections Spatial positioning and translation and Fatigue crack quantification method).

We built the mechatronic architecture around an STM32 microcontroller, which directly controls the magnetic wheels, arm servos, and localization sensors, as shown in Fig. 2d. A Raspberry Pi operated in parallel as the controller for image acquisition and data processing. Independent dual power systems (24 V and 12 V lithium batteries) ensured a stable supply for both drive and electronics. Modules communicate via a local Wi-Fi access point, enabling real-time data transmission. Sensor placement was optimized for spatial consistency: encoders were embedded in the wheel shafts, laser sensors were aligned with the chassis axis, and the IMU was mounted parallel to the chassis plane (x-axis aligned with forward motion).

Spatial positioning and translation

Precise localization inside electromagnetically shielded steel box girders is essential to assign cracks to their structural domains. We first analyzed positional requirements in three orthogonal directions: transverse, longitudinal, and vertical. Transverse accuracy needed only to distinguish welds separated by >300 mm; longitudinal accuracy was critical for deck-rib welds due to their visual similarity; and vertical accuracy was needed to differentiate deck-rib from diaphragm-rib cracks, which differ in elevation by >200 mm.

For the closed electromagnetically shielded steel box girder environment, we selected a combined positioning system consisting of a UWB sensor, an IMU sensor, and a pair of laser sensors. We achieved localization inside the steel box girder by fusing data from an IMU, a UWB sensor, and forward/rear laser rangefinders using a factor graph approach (Fig. 3a). The UWB sensor provides the robot's 3D position (x, y, z), the IMU yields orientation and acceleration, and the lasers measure distances to the girder walls. The UWB sensor measures the robot's local spatial coordinates $[x, y, z]$ within the steel box girder. The IMU sensor measures the robot's angles $[\theta_{yaw}, \theta_{roll}, \theta_{pitch}]$, accelerations $[a_x, a_y, a_z]$, and angular accelerations $[\beta_x, \beta_y, \beta_z]$ during motion. Forward and backward laser sensors measure the distance from the robot to the boundary $[d_{front}, d_{rear}]$.

We established factor nodes for the three sensors and used a factor graph approach to fuse multi-sensor measurement data, constructing a unified factor graph model. This model uses the robot's poses $[x, y, z, \theta_{yaw}, \theta_{roll}, \theta_{pitch}]$ at different moments as optimization variables. By maximizing the joint maximum a posteriori probability of all factor nodes, a nonlinear optimization algorithm was used to solve the most likely global robot pose sequence in real time. GTSAM was used to implement factor graph multi-sensor fusion, and localization results were output at a 2 Hz frequency. Since the robot's inspection strategy can be divided into a diaphragm inspection phase and a roof inspection phase, the most prominent measurement difference between the two phases is the IMU pitch angle. Therefore, the IMU pitch angle measurement was used to determine the inspection phase. This provided a global pose sequence while mitigating drift. The IMU pitch angle further identified inspection phases (deck or diaphragm). By identifying which structural segment (deck or diaphragm) the robot is in (via IMU pitch angle and coordinates, Fig. 3b), the system can tolerate slightly larger positioning errors at that module level while still accurately assigning each crack to the correct weld line. Each crack was ultimately assigned to a specific weld line, with its position translated into

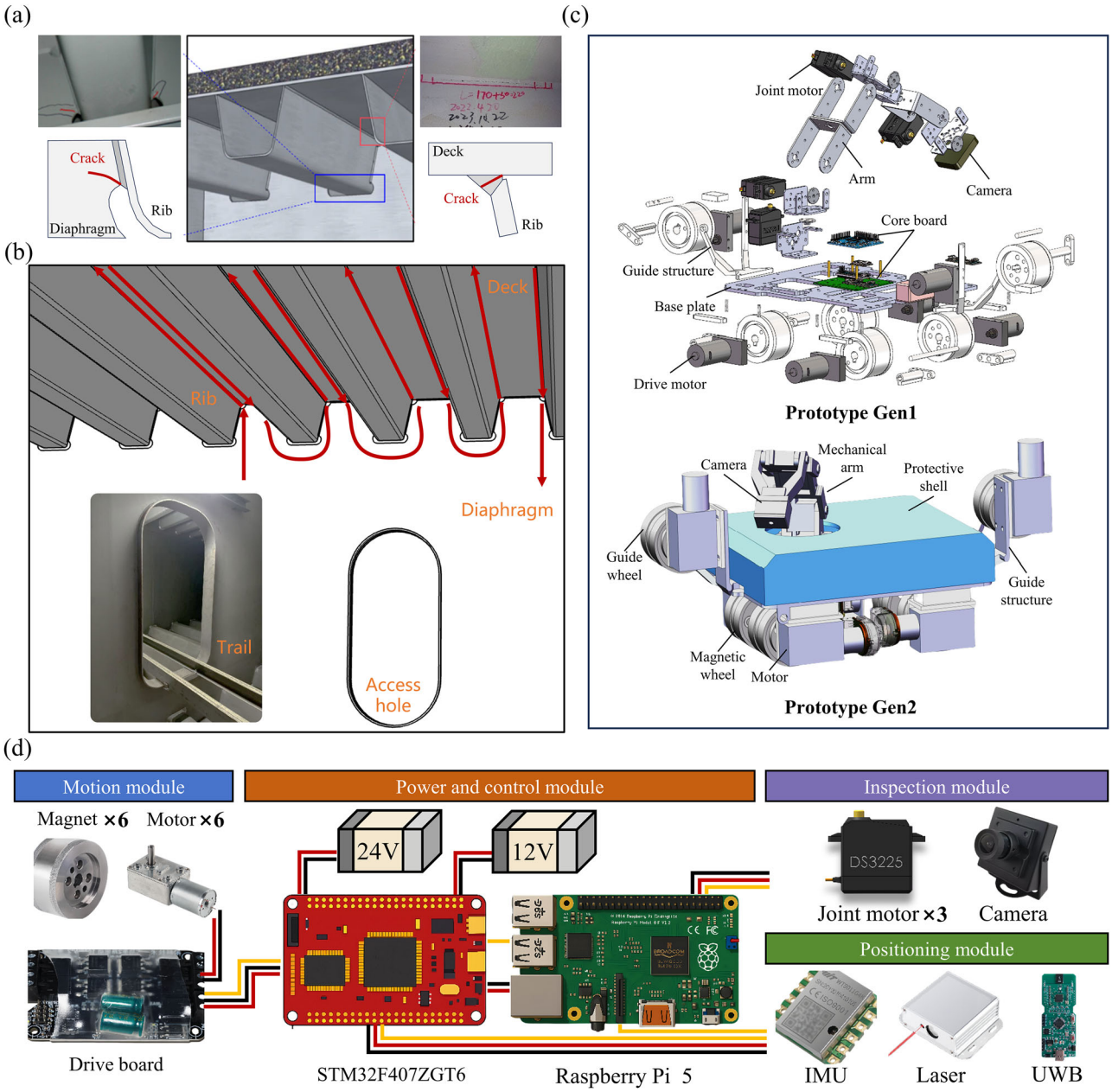


Fig. 2 | Inspection strategy of fatigue cracks in OSDs. a Fatigue crack distribution. **b** Inspection route. **c** Configuration of the inspection robot. **d** Design of integrated mechatronic systems.

descriptive natural language relative to rib or diaphragm indices and distances, as shown in Fig. 3c. In the laboratory validation, the error statistics were computed against a marker-based ground-truth baseline constructed from calibrated fiducials, with IMU-verified quasi-constant-velocity motion used to support interpolation between checkpoints.

Fatigue crack quantification method

Mapping crack pixels to global coordinates required calibration of the robotic arm and onboard camera, as shown in Fig. 4a. First, the transformation from the robot base to the arm end-effector was solved using forward kinematics based on Denavit-Hartenberg (DH) parameters[41]. The camera, rigidly mounted to the arm, was related to the end-effector through a translation matrix. Next, Zhang's checkerboard method was used to calibrate intrinsic and extrinsic parameters of the RGB camera[42], linking image pixel coordinates (u, v) to camera coordinates (xc, yc, zc). By further applying rotation and translation matrices, cracks in image space were mapped into global structural coordinates. For the deck-rib welded joint, the acquisition direction is constrained to be approximately perpendicular to the weld. This constraint provides a consistent external reference axis (the weld direction) to define the image coordinate origin and thus localize cracks, given the limited visual references at this detail. For diaphragm-rib cracks, the captured images typically contain sufficient reference features to define the image origin reliably; therefore, the same perpendicular-to-weld constraint is not required to determine crack coordinates.

The robotic arm's camera captured high-resolution weld images during inspection, as shown in Fig. 4a. We identified and quantified cracks using an intermediate-supervision segmentation network (IS-Net), which integrates a ground-truth encoder and a nested segmentation module derived from U^2-Net. IS-Net improves sensitivity to fine-scale cracks and preserves performance on high-resolution images, as shown in Fig. 4b.

A dataset of 5200 fatigue crack images (4032 × 3024 pixels), collected from a steel bridge in service for over 15 years, was used for training, validation, and testing (7:2:1 ratio). Two representative crack types were included: (i) deck-rib weld cracks and (ii) diaphragm-rib weld cracks. After

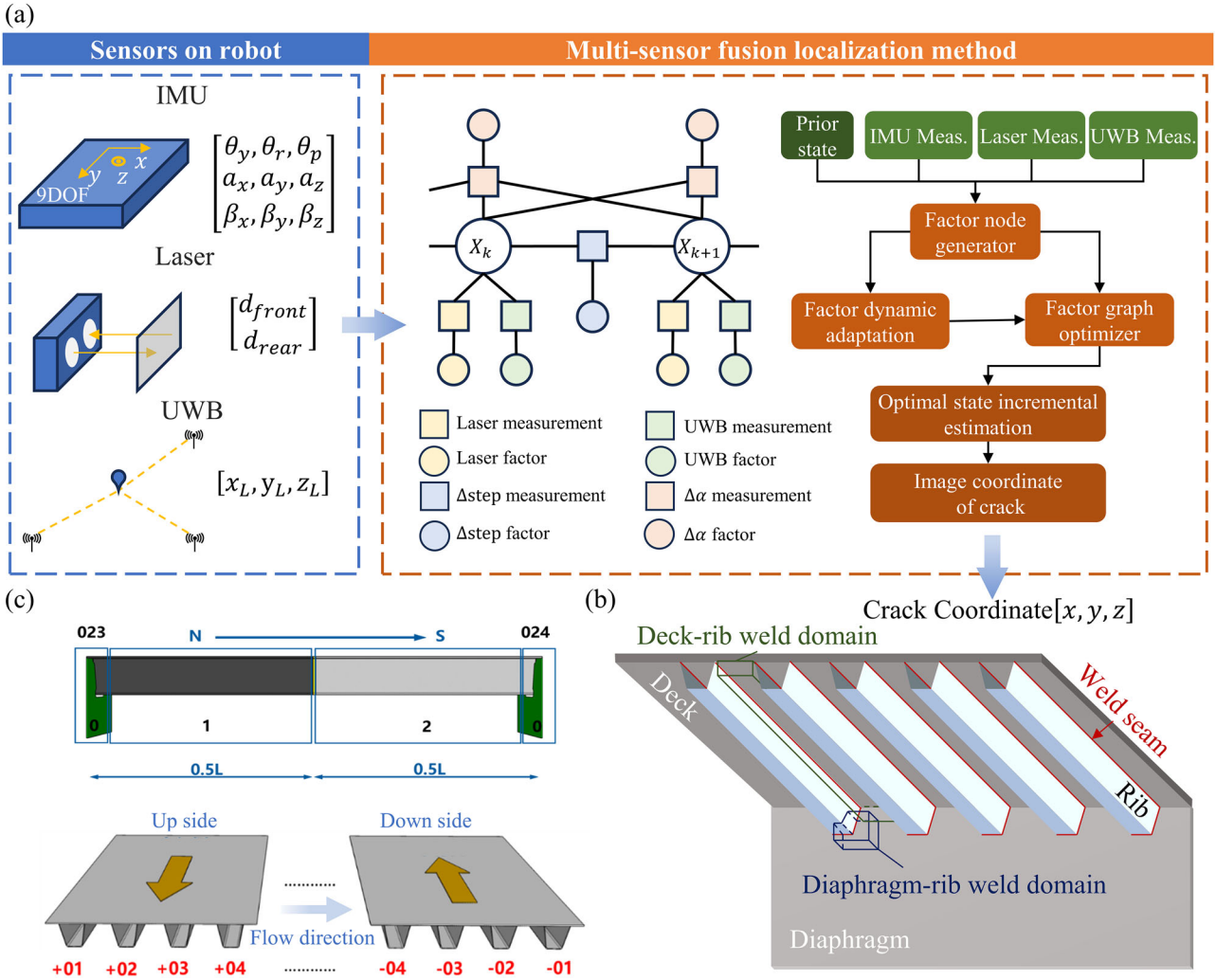


Fig. 3 | Spatial positioning system. a Principle of the positioning system. **b** Domain segmentation of the weld joints in OSDs. **c** Nature expression of the crack code.

hyperparameter tuning, IS-Net was trained until convergence, yielding a model deployed for real-time inference onboard the robot. We introduced a cosine annealing schedule with a warm-restart technique to stabilize training. Several key parameters were preset: warmup_factor=1×10^{-3}, warmup_epochs=1, and end_factor=1×10^{-6}. The remaining hyperparameters were set as follows: batch size=8, initial learning rate=1×10^{-3}, weight decay=1×10^{-4}, and epochs=300.

During inspection, captured images were processed to generate crack segmentation masks, which were refined using morphological opening/closing operations, as shown in Fig. 4c. Skeletonization was then applied to extract the crack centerline. To enable accurate FE insertion, a piecewise insertion method was adopted: skeleton lines were divided at 100-pixel intervals, each segment endpoint was transformed from pixel to global coordinates, and ordered nodes were subsequently imported into the FE model.

DT model updating

The DT was updated by embedding observed cracks into a high-fidelity FE model of the OSDs. The model geometry was based on a half-span segment of an in-service cable-stayed bridge, with boundary constraints applied at the tower base and beam ends, as shown in Fig. 5a. Material properties corresponded to Q345qD steel (elastic modulus 2.06 × 10^5 MPa, Poisson's ratio 0.3). Before crack insertion, the FE model uses a locally refined mesh around the weld region with an element size of 1 mm. After crack insertion, FRANC3D automatically refines the mesh and performs local remeshing around the crack front at each propagation increment, further refining the discretization in the vicinity of the crack front. The applicability of the FRANC3D-based crack-growth simulation for fatigue cracks in OSD welded joints has been validated in our previous work[43].

For thin-walled steel structures of OSDs, the deck-rib cracks initiate at the weld root and extend to the weld surface[44], and the diaphragm-rib cracks are usually through-going cracks[45]. Therefore, the surface two-dimensional cracks are reconstructed into three-dimensional cracks, as shown in Fig. 5b. We reconstructed crack surfaces from skeleton lines. For diaphragm-through cracks, nodes were projected to the opposite surface and connected into a continuous crack face. The continuous crack face is a reconstructed geometric representation obtained by interpolating the discrete crack-front points extracted from inspection data. For deck-rib weld cracks, skeletons were projected to the weld root and connected. FRANC3D was then used to remesh crack regions with quarter-point wedge elements near crack tips and brick/pyramid transitions for compatibility. This procedure enabled realistic crack-insertion and propagation analysis.

Fatigue performance evaluation

We developed a closed-loop fatigue performance evaluation process that begins with robotic inspections of in-service steel bridges to map the locations and sizes of GT cracks. These observed cracks are then embedded into a high-fidelity FE model of the bridge. The inspection-derived crack geometry and location are imported via scripts for coordinate mapping and crack-surface generation (with local remeshing in FRANC3D). Using the

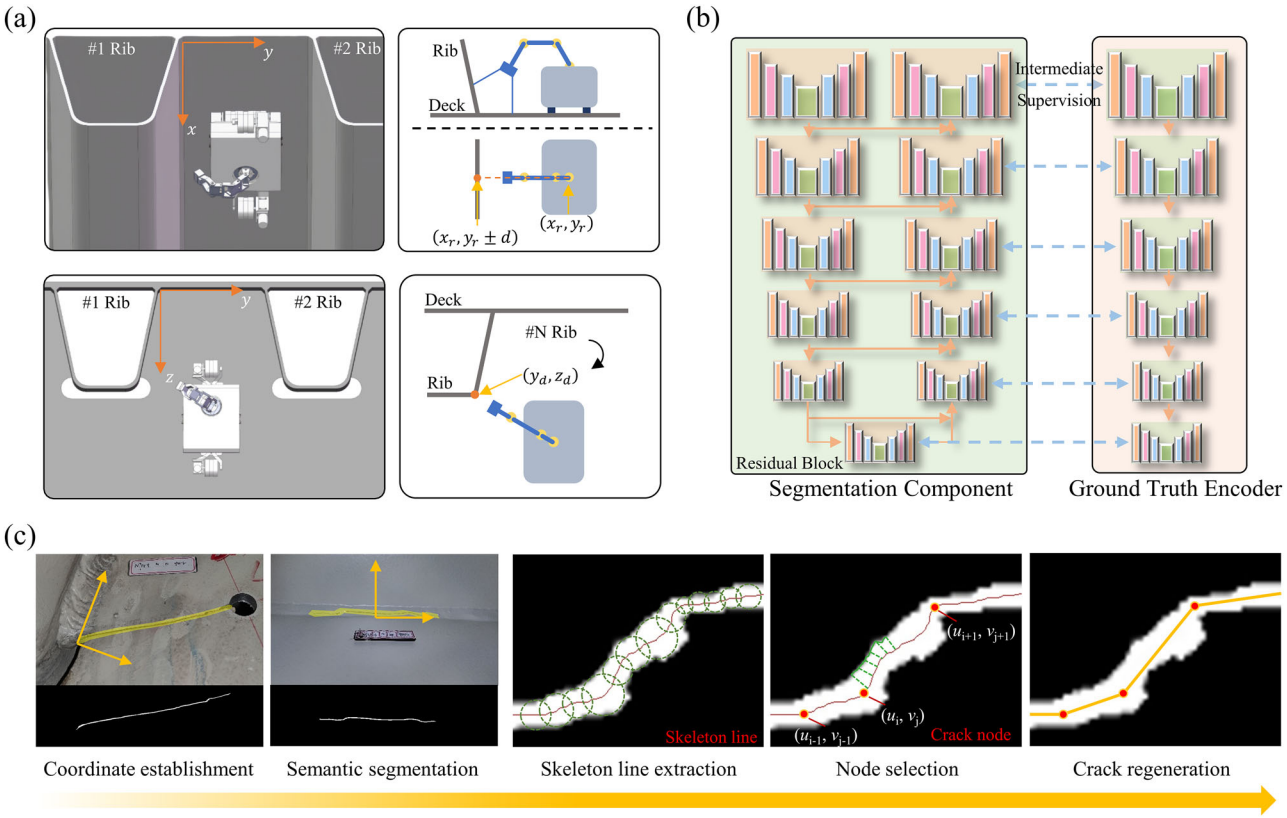


Fig. 4 | Fatigue crack quantification method and local location registration. a Crack image capture method. **b** Semantic segmentation model. **c** Extraction of fatigue crack skeleton lines and coordinate establishment.

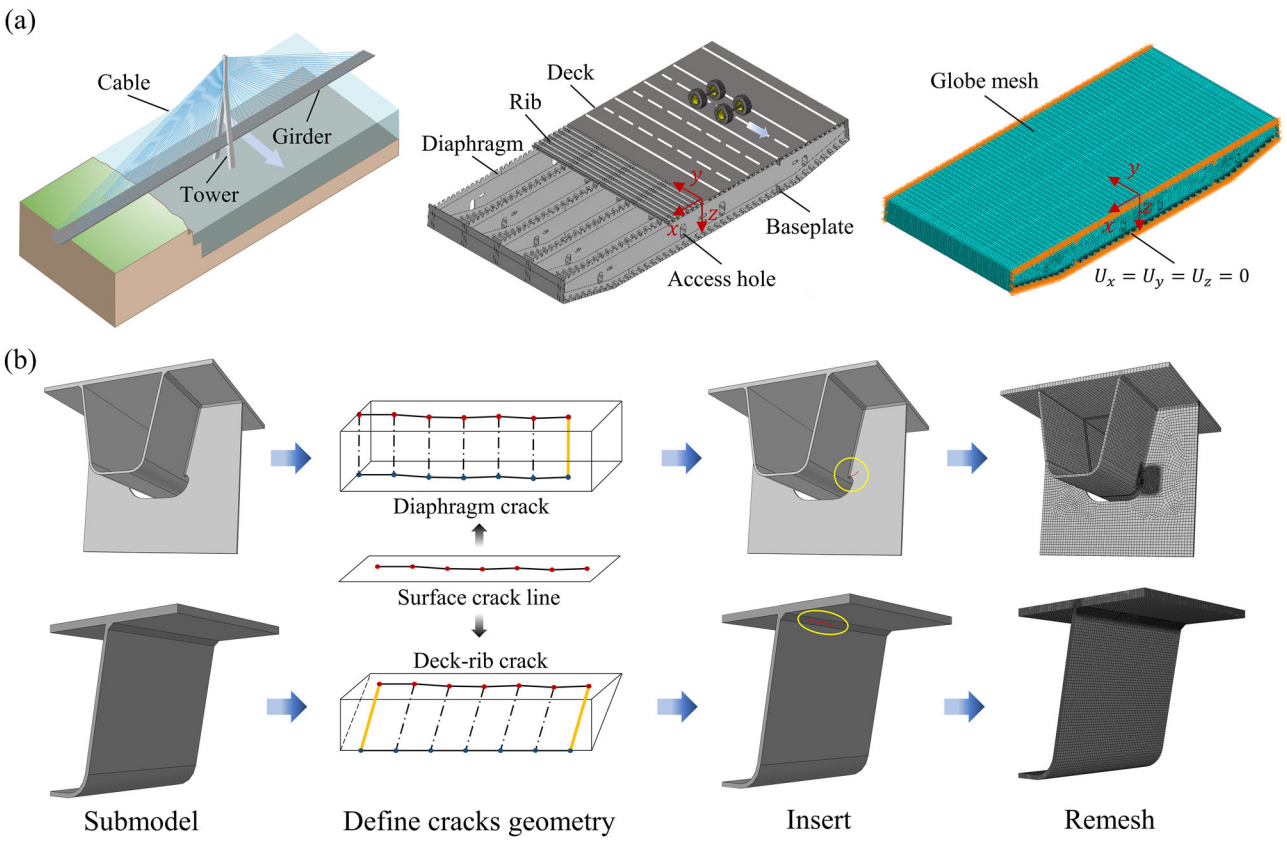


Fig. 5 | DT Model Update Strategy. a Finite element bridge model. **b** Crack insertion and mesh regeneration.

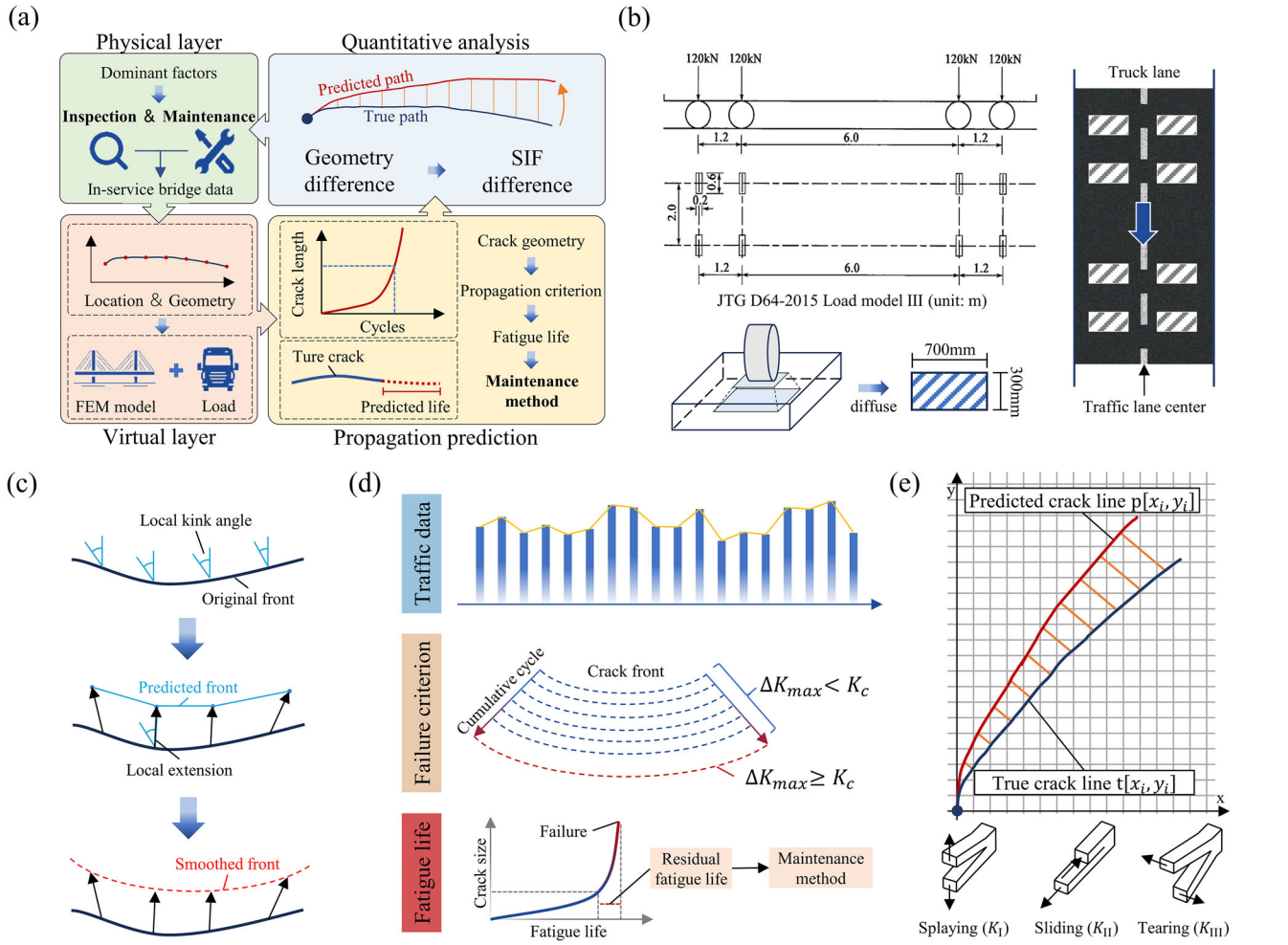


Fig. 6 | Fatigue performance evaluation. a Evaluation loop. **b** Load scheme. **c** Crack propagation method. **d** Fatigue life evaluation. **e** Difference quantization.

applied vehicle load spectrum, the model predicts crack propagation and remaining life, which informs maintenance decisions. Furthermore, discrepancies between the predicted and observed crack growth are analyzed to update the model. By quantifying differences in crack path via a crack path deviation function, we can identify key factors to improve the DT's predictive accuracy (Fig. 6a).

Traffic loading was applied following the Chinese code JTG D64-2015 (model III)[46], equivalent to dual-axle wheel loads on deck plates, as shown in Fig. 6b. A 120 kN wheel load was distributed over an effective area of 300 × 700 mm. Traffic loading was applied in the bridge's heavy-traffic lane and incrementally shifted along the span in 200 mm intervals.

Stress intensity factors (SIFs) at crack fronts were computed using the M-integral[47], as shown in Fig. 6c. Crack propagation was simulated in FRANC3D using the NASGRO model[43], which accounts for stress ratio R and crack closure effects. The crack growth rate was described as a function of SIF range and fracture toughness, while propagation direction was determined by the maximum energy release rate criterion. This enabled three-dimensional crack evolution under mixed-mode conditions. The computed SIFs were verified to be mesh-insensitive within 3% when halving the local mesh size. Using the DT model's crack growth simulation results and standard fatigue failure criteria, we predicted the remaining fatigue life of the in-service OSD (Fig. 6d).

We quantitatively compared each DT crack path to its corresponding OBS crack (Fig. 6e). The comparison metrics included the differences in node-to-node distances and in crack deflection angles. The nodes on the OBS and the DT crack are defined as:

$$P = \{p_i\}_{i=1}^{N_P} \subset R^3$$
$$O = \{o_j\}_{j=1}^{N_O} \subset R^3 \quad (1)$$

Calculate the nearest-neighbor distance on two crack. For each predicted point p_i, we define the closest-observation distance:

$$d_i = \min_{1 \le j \le N_O} ||p_i - o_j||_2 \quad (2)$$

The dist-ance between nodes P and O is described by the Bidirectional Hausdorff distance[48]:

$$H(P, O) = \sup_{p \in P} \inf_{o \in O} ||p - o||_2$$
$$H_{sym}(P, O) = \max(H(P, O), H(O, P)) \quad (3)$$

the Chamfer distance reflects the overall geometric deviation:

$$D_{Chamfer}(P, O) = \sqrt{\frac{1}{2}\left(\frac{1}{N_P}\sum_{i=1}^{N_P}\left(\min_j ||p_i - o_j||\right)^2 + \frac{1}{N_O}\sum_{j=1}^{N_O}\left(\min_i ||o_i - p_j||_2\right)^2\right)} \quad (4)$$

angular deviation is quantified by the local deviation angle:

$$\theta_k = \arccos\left(\frac{\tau_k^{\text{pred}} \cdot \tau_{m(k)}^{\text{obs}}}{||\tau_k^{pred}||_2||\tau_{m(k)}^{obs}||_2}\right) \quad (5)$$

Morphological validation was performed by comparing DT crack skeletons with observed ones. Deviations were quantified by Hausdorff distance, and local orientation differences were expressed as deflection angles. These discrepancies were interpreted in terms of mode I, II, and III SIF differences, enabling back-analysis of underrepresented shear or torsional effects in the model.

Crack growth behavior is a key issue in damage assessment. The crack propagation morphology is fundamentally determined by the stress intensity factor (SIF)[49]. For a fatigue crack in a given structure, the SIFs can be calculated for the corresponding load condition and used to predict the crack growth rate and direction. This is also the main principle of current fatigue crack growth simulation. For cracks subjected to mixed fatigue loading, the growth rate is controlled by the equivalent SIFs KI, II, and III, as shown in the following formula[50]:

$$\Delta K_v = \frac{\Delta K_{\text{I}}}{2} + \frac{1}{2}\sqrt{\Delta K_{\text{I}}^2 + 5.3361\Delta K_{\text{II}}^2 + 4\Delta K_{\text{III}}^2} \quad (6)$$

Since Mode I generally dominates the total stress-intensity field, $K_{\text{I}}, K_{\text{II}}$ and K_{III} can be stably calculated to predict crack growth rate. However, crack deflection is primarily governed by the fluctuating K_{II} component.

$$\varphi = \mp\left[140 \times \frac{|K_{\text{II}}|}{K_{\text{I}} + |K_{\text{II}}| + |K_{\text{III}}|} - 70 \times \left(\frac{|K_{\text{II}}|}{K_{\text{I}} + |K_{\text{II}}| + |K_{\text{III}}|}\right)^2\right] \quad (7)$$

During the simulation process, the difference in SIFs at key crack node locations can be calculated. Using the OBS crack morphology as a benchmark, the DT model results can be used to analyze SIFs at locations where the simulated and OBS cracks coincide and to predict subsequent expansion behavior. For example, if there is a difference in crack deflection angle, the difference in SIFs can be deduced using the formula. Instead of using the SIF differences to correct or recalibrate the model, the mixed-mode SIF ratio is interpreted here strictly as a diagnostic indicator that reveals where the DT prediction becomes mechanically unreliable under in-service mixed-mode loading conditions. The observed morphology deviation is therefore not used to modify the current DT formulation, but to expose the intrinsic limits of crack-propagation predictability when shear-dominated fracture behavior emerges.

Results

Laboratory validation of robotic inspection

A laboratory-scale OSDs segment containing three closed U-ribs was fabricated to verify the fundamental functionality of the proposed robotic inspection system, as shown in Fig. 7a. The specimen geometry and rib spacing were designed to replicate those of an in-service steel bridge, ensuring representativeness of the validation tests. The inspection paths reproduced the actual field sequence, starting from diaphragm deployment and transitioning to deck plate inspection, thereby preserving operational realism.

We first characterized the adhesion performance of the magnetic wheels under conditions representative of decks and diaphragms, as shown in Fig. 7b. We measured adhesion strength using a coating thickness gauge (0-1800 μm, resolution 1 μm, error ±3%) and a tensile tester(50-500 N, resolution 0.1 N, error ±1%). As shown in Fig. 7c, the adhesion force exhibited an inverse cubic relation with distance, with slightly higher pull-off forces on diaphragms than on decks. Overall, each wheel provided an additional adhesion capacity exceeding 80 N, sufficient to guarantee stable locomotion even on vertical or horizontal surfaces. To assess whether the measured adhesion is sufficient for field operations, we performed a conservative margin check. The robot mass is 4.5 kg ($W = mg = 44.1N$). The most unfavorable condition occurs during wall-to-wall transitions, where only three magnetic wheels remain in contact. With a lower-bound adhesion of 80 N per wheel, the available holding force is $F_{adh,\min} = 240N$. Introducing a conservative dynamic amplification factor $\gamma = 2$ to account for vehicle-induced vibrations, coating variability, and local deformations, the safety criterion $F_{adh,\min} \geq \gamma W$, corresponding to a safety factor of 2.72 in the worst-case contact state (and 5.44 when all six wheels are engaged). In addition, coating-thickness measurements on the in-service bridge indicate $183 \pm 21\mu m$ on the deck and $203 \pm 36\mu m$ on the diaphragm; within this coating range, pull-off tests confirm that each wheel maintains an adhesion force of at least 80 N.

Subsequent mobility tests (Fig. 7d) demonstrated that the robot could traverse the OSDs segment smoothly and transition reliably between decks and diaphragms. During locomotion, the onboard robotic arm adjusted its orientation to capture weld images using the high-resolution camera. The acquired images (Fig. 7e) preserved weld bead details with clarity, confirming that the system is capable of resolving fatigue cracks at the sub-millimeter scale.

Figure 7f shows the positioning error, defined as the root-mean-square difference between the robot's measured position and its ground-truth baseline (Eq. 1) :

$$E = \sqrt{\sum (L_m - L_t)^2} \quad (8)$$

where, L_m is the measurement position and L_t is the real position of the robot location.

In the test, fiducial markers were used as external geometric references to establish a specimen-fixed coordinate and to register reference positions along the predefined inspection path. The onboard IMU verified quasi-constant-velocity motion (measured longitudinal acceleration close to zero). Under the marker constraints and the verified motion state, the trajectory of the designed path in the experimental coordinate system was treated as a ground-truth baseline for accuracy assessment. The positioning error was computed as the RMS difference between the multi-sensor localization output and the baseline. According to the error statistics of the positioning system in the test (Fig. 7f and g), the average positioning errors in the y and z directions are both under 2 cm. The error variability is well-controlled, with a maximum error of under 5 cm. Given that the spacing between weld lines exceeds 200 mm, these accuracies are sufficient for domain-level crack assignment and meet the requirements of structural state updating.

Because realistic fatigue cracks could not be introduced into the specimen, the full closed-loop workflow was not executed in the laboratory. Instead, the tests focused on verifying key functionalities—adhesion, mobility, imaging, and localization—that underpin the overall system. These results establish that the robotic platform achieves the mechanical robustness and sensing accuracy necessary for deployment in real bridge conditions, providing the technical foundation for subsequent field implementation.

Implementation on an in-service steel bridge

To demonstrate applicability under real conditions, the framework was deployed on a cable-stayed bridge in China that has been in service for over 15 years, as shown in Fig. 8a. Based on the comparative analysis of OBS measurements and DT cracks for the two types of fatigue cracks in the test set, the error distribution approximately follows a normal distribution. At the 95% confidence level, the length error interval is [−0.8 mm, +0.7 mm]. Results show that 92.3% of the samples deviate less than 10% from the true values, confirming that the method meets practical engineering accuracy requirements. The robotic inspection unit performed five independent scans inside a steel box girder and identified two representative fatigue cracks: diaphragm-rib joint and deck-rib joint. Each automated inspection

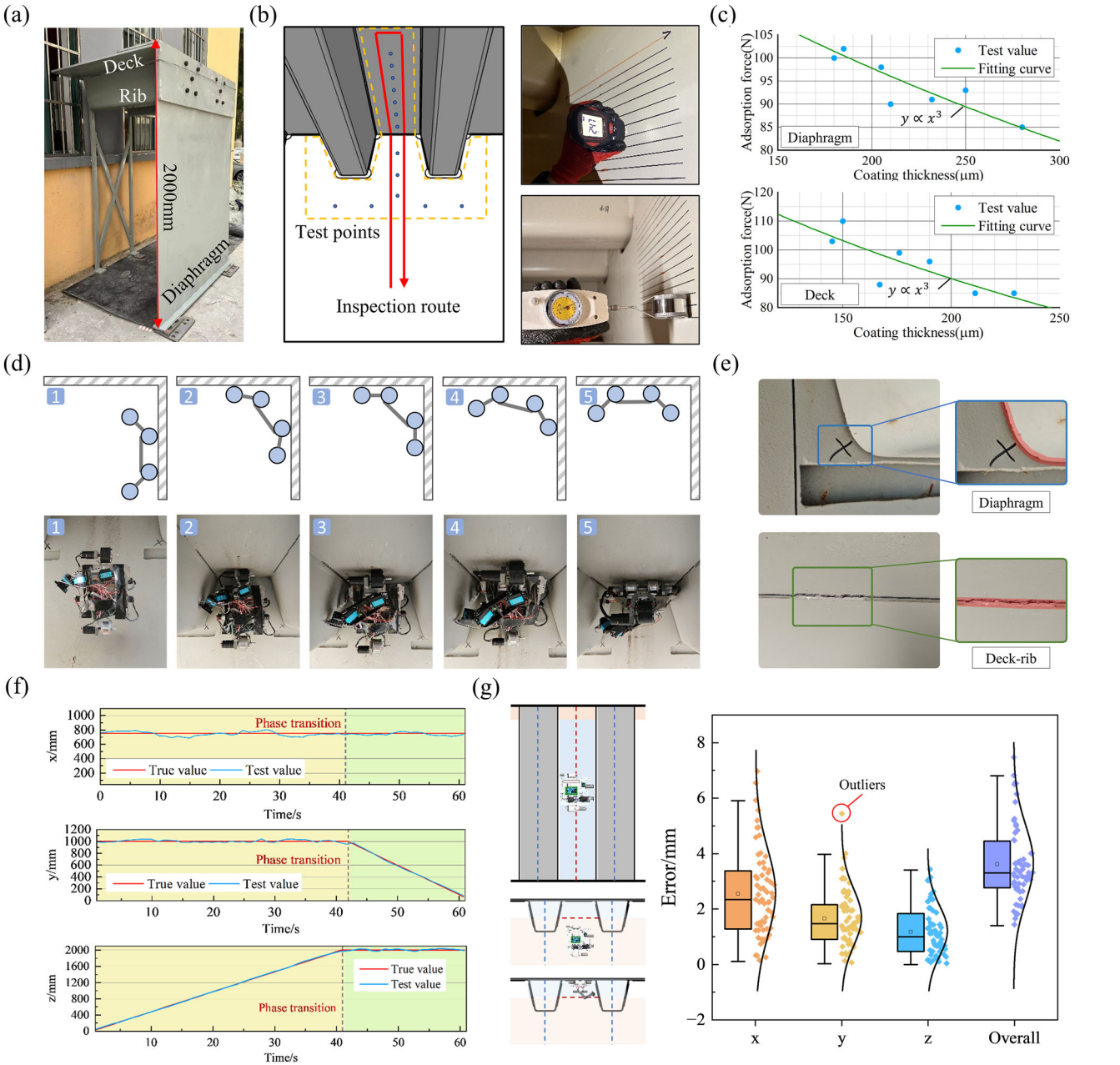


Fig. 7 | Functional experiments and results in laboratory. a Experimental environment. **b** Inspection path and adhesion force test. **c** Pull-off force of different plates. **d** Motion stage of bridge inspection. **e** Crack image captured by camera. **f** Location in the motion process. **g** Location error.

cycle required on average 50.4 min ($N = 5$), versus about 125 min for manual inspection—a 59.6% reduction in inspection time. Across the five independent scans, no wheel slippage or detachment was observed, indicating stable adhesion under in-situ vibration and surface conditions, as shown in Fig. 8b. Manual duration is inherently variable across inspectors; therefore, the reported manual time is intended to represent the average working time of experienced inspectors under typical practice.

To assess the performance of the proposed algorithm, we trained several state-of-the-art semantic segmentation models on two representative fatigue crack images. The corresponding inference results are shown in Fig. 9. Qualitatively, IS-Net achieved the best performance on both crack types. Compared to the three other segmentation models, IS-Net's intermediate supervision mechanism enabled more precise delineation of fine crack details and reduced misclassification.

Each observed crack skeleton was incorporated into a high-fidelity finite-element model and propagated using FRANC3D under the JTG D64-2015 Model III load spectrum[46]. To quantitatively evaluate the agreement between simulated and observed cracks, we computed morphological metrics including the bidirectional Hausdorff distance, the Chamfer distance, and the local deflection angle along the crack.

For diaphragm-rib crack, Fig. 10a shows the DT model update process for the collected cracks. Figure 10b compares the measured crack morphology with the DT crack morphology. After 25 mm simulated propagation, the DT crack path matched the OBS crack morphology with a maximum point-to-point deviation below 3 mm (Fig. 10c). These results indicate that, for near-surface diaphragm-rib cracks, the DT captures both the curvature and the propagation direction with millimeter-scale precision. For the diaphragm-rib crack, the crack propagation is primarily driven by out-of-plane deformation of the diaphragm under structural constraints and external loading[51]. As a result, the crack exhibits mixed-mode fatigue behavior, leading to relatively large K_I and K_{II} (Fig. 10d). In addition, the stress intensity factor of the crack front that grows through the diaphragm is

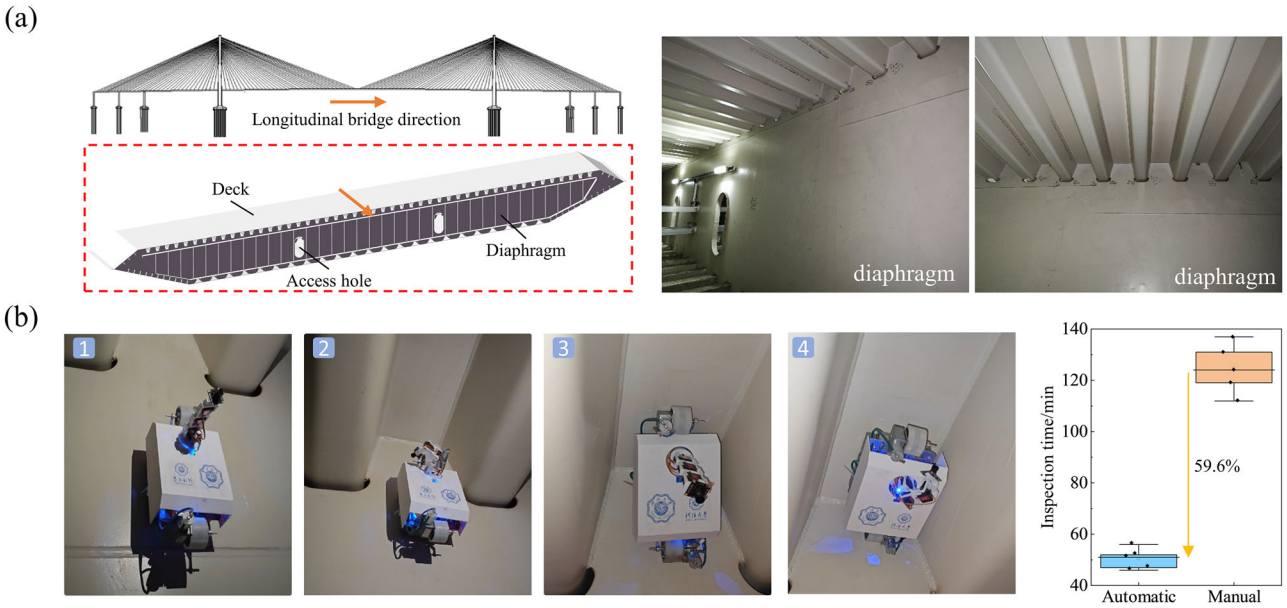


Fig. 8 | Experimental details in in-service bridge. a Experimental environment of in-service bridge. **b** Motion process of inspection strategy.

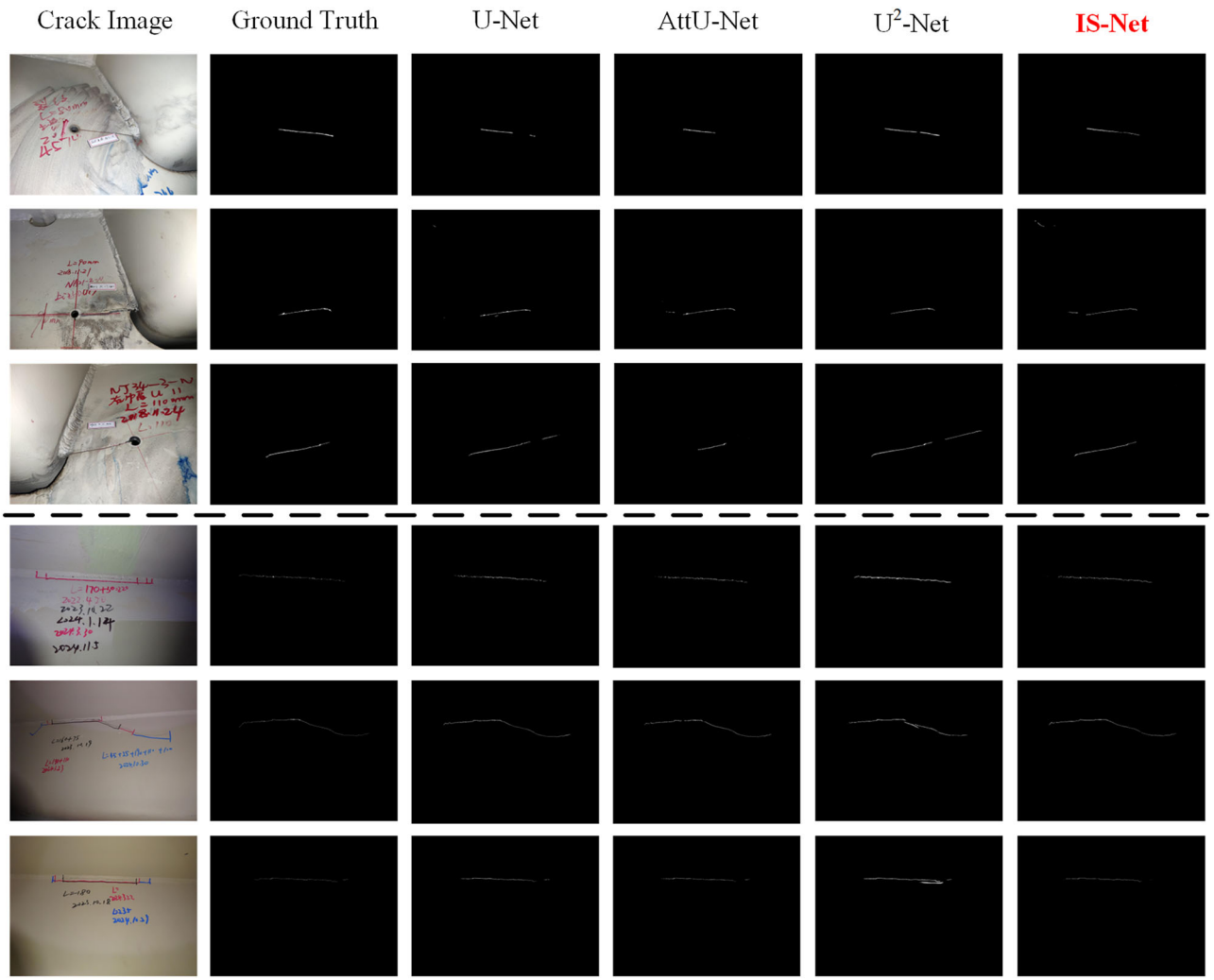


Fig. 9 | Qualitative comparison of semantic segmentation results for fatigue cracks using U-Net, AttU-Net, U²-Net, and IS-Net models. The approach involves training several state-of-the-art semantic segmentation models on representative fatigue crack images, and comparatively analyzing their inference results to evaluate segmentation performance, thereby identifying the most effective algorithm for crack detection.

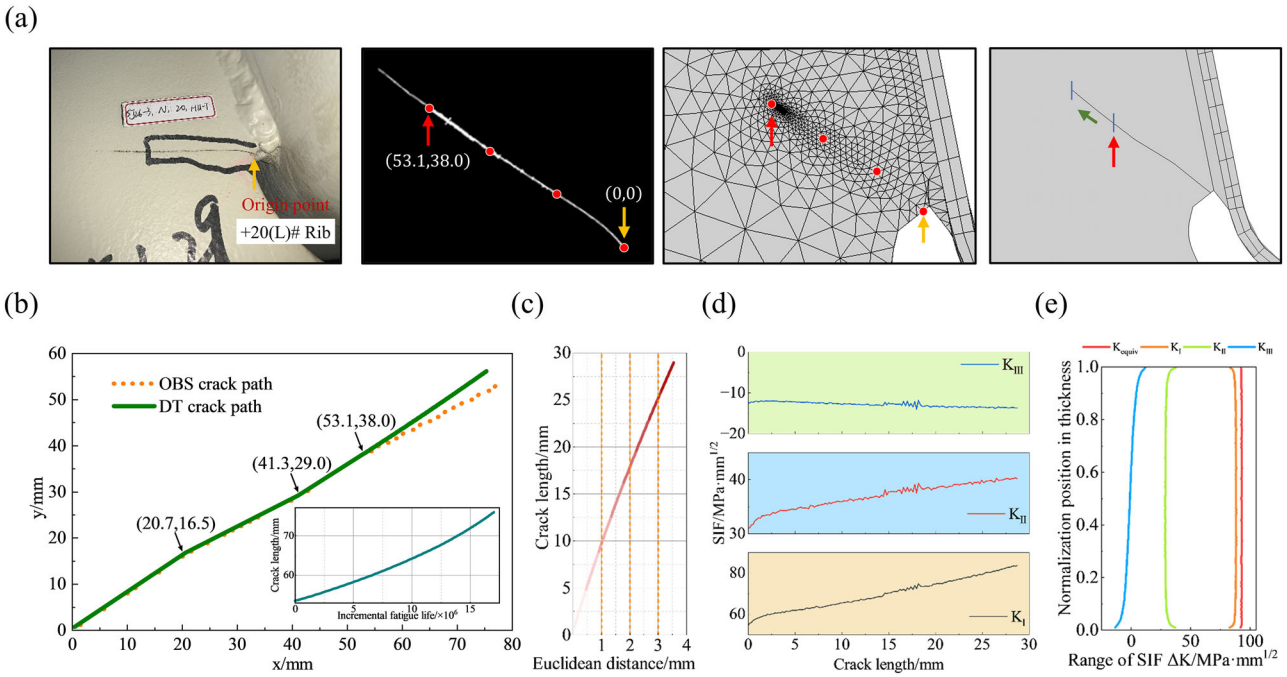


Fig. 10 | DT-based analysis results of diaphragm-rib weld joint cracks in in-service bridge. a DT model update process. **b** Crack morphology. **c** Deviation between DT cracks and OBS cracks. **d** Stress intensity factor in crack propagation. **e** SIF distribution along crack front.

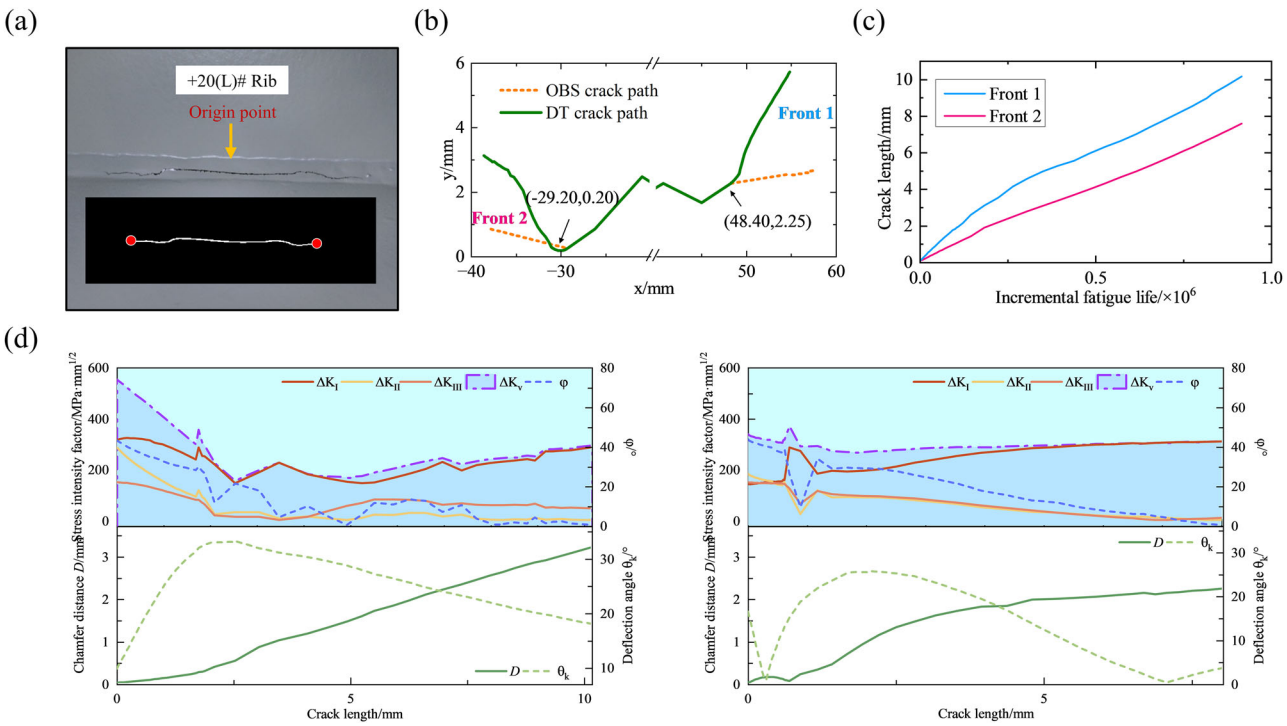


Fig. 11 | DT-based analysis results of deck-rib weld joint crack s in in-service bridge. a DT and OBS crack skeleton. **b** Crack morphology. **c** Number of cycles versus crack length curve. **d** Stress intensity factors and morphological differences.

extracted, as shown in the Fig. 10e. It can be seen from the figure that the stress intensity factor of the crack front is relatively balanced in the thickness direction. As the crack length increases, the crack's growth rate also increases, causing the crack to expand faster. Manual inspection frequency is often limited by labor and access constraints, cracks may be discovered at a relatively advanced stage; the proposed robotic inspection can increase inspection frequency and shorten detection latency, enabling earlier identification of near-critical cracks and more timely maintenance.

In contrast, deck-rib crack exhibited larger geometric deviations. It has two extension fronts, as shown in Fig.11a. The two fronts are defined as front 1 and front 2. The predicted results are compared with the OBS crack morphology, as shown in Fig. 11b. Front 1 exhibits a faster growth rate than Front 2, consistent with field observations of such cracks (Fig. 11c). To explore the mechanical source of this mismatch, we analyzed the SIF outputs from FRANC3D. For each propagation step, nodewise K_I, K_{II}, and K_{III} values were extracted, and the propagation rate ΔK_v and crack deflection φ

were computed to describe the local mixed-mode dominance. We then correlated the local morphology deviation with the SIF parameters across all front nodes. Figure 11d presents the evolution of K_I, K_{II}, and K_{III} distributions along the crack propagation for both cracks. Front 1 shows higher spatial variability in K_{II} and greater local peaks in crack deflection φ, coinciding with the observed deflection zones. Figure 11d further demonstrates that the mean Hausdorff distance increases linearly with propagation length for front 1, while more stable for front 2. Fatigue-life estimation based on FRANC3D-derived SIF histories and the NASGRO equation yielded remaining equivalent load cycles for front 1 and front 2. Front 1, which exhibits higher deflection φ variability and larger geometric deviations, should therefore be prioritized for short-interval follow-up inspections. The DT crack-front geometry deviated from the observed crack by a Hausdorff distance of 3.19 mm for front 1 and 2.22 mm for front 2. The predicted mean deflection angle was 23.5 ± 6.2 for front 1 and 10.7 ± 7.9 for front 2, in comparison to the field-measured crack. The deviation between the DT crack and the OBS crack increased linearly with crack length. Compared to diaphragm-rib weld cracks, deck-rib weld cracks exhibit a larger deviation between predicted and OBS crack paths, which can lead to greater errors in the deflection angle prediction. A strong positive correlation was found between crack deflection φ and K_{II} for front 1. These results show that a higher Mode-II contribution is associated with larger morphology deviations, indicating lower prediction confidence under mixed-mode-dominant conditions.

Overall, by combining crack-front morphology and SIF fields from FRANC3D, we show that (i) the DT reproduces observed crack shapes within 10 mm accuracy for single-front, near-surface cracks; (ii) because morphology deviation increases with Mode-II contribution, the local $|K_{II}/K_I|$ ratio can be used as a quantitative indicator of predictive uncertainty along the crack front.

Discussion

The present study establishes a quantitative framework to evaluate the predictive accuracy of DT for fatigue cracks in in-service steel bridges by integrating robotic inspection, high-fidelity FEM, and FRANC3D-based crack-propagation analysis. The reduced per-cycle time enables more frequent, risk-informed inspections that can be integrated into the standard inspection-assessment-maintenance workflow: crack geometry acquired at fatigue-critical details is used to update the DT, prioritize follow-up inspections and repairs, and tighten inspection intervals when rapid growth is indicated. DT updating is currently semi-automated, with inspection-derived crack geometry imported via scripts and minor manual checks; a ~ 10 mm propagation update typically requires 3-10 hours depending on model size and computing resources. This closed-loop process shortens detection latency and projects interval growth, reducing the risk of cracks growing unnoticed between scheduled inspections. In the tested cases, the DT reproduced crack morphologies with millimeter-level precision and provided physically interpretable uncertainty indicators, though broader validation across more bridges is still needed.

We found a strong link between morphological deviation and mixed-mode fracture behavior. Importantly, this relationship is not used as a correction mechanism for the model, but as a physics-based indicator of where DT predictions lose reliability under mixed-mode dominance. For the near-surface diaphragm-rib crack, where Mode I dominates, the DT achieved a maximum point-to-point deviation below 3 mm and a nearly constant Hausdorff distance over 25 mm propagation. In contrast, for the deck-rib crack, characterized by higher shear contributions, the deviation increased proportionally with propagation length and correlated positively with the local ratio $R_{II/I} = |K_{II}|/|K_I|$. This quantitative relationship reveals that the Mode-II component of the stress-intensity field governs the local curvature and directional uncertainty of crack growth. In other words, mixed-mode dominance is strongly correlated with DT predictive accuracy.

The morphology-SIF correlation offers a mechanistic explanation for the limits of DT fidelity. In Mode-I-dominated conditions, crack propagation follows a stable, energetically preferred plane, and the DT—built on linear elastic fracture mechanics—accurately tracks both geometry and stress field evolution. However, under mixed-mode loading, small perturbations in local shear stresses or weld geometry amplify K_{II} fluctuations, leading to bifurcations in the crack front and greater geometric dispersion. The observed proportionality between $R_{II/I}$ and the Hausdorff deviation thus provides a physics-based measure of DT reliability without invoking additional empirical calibration.

Our findings offer broader implications for developing self-assessing DTs. By continuously tracking the evolution of $R_{II/I}$ and its correlation with predicted morphology, the DT can autonomously identify zones of low predictive confidence. This capability transforms the DT from a purely forward-predictive tool into a diagnostic system capable of adaptive calibration and targeted monitoring.

In summary, the proposed morphology-SIF analysis quantitatively links mixed-mode fracture mechanics to DT reliability, providing both a diagnostic foundation and a pathway toward self-evaluating DTs for infrastructure life-cycle management.

Future work

Despite these advances, several limitations should be acknowledged. The field validation was conducted on two representative cracks due to accessibility constraints. Although these cases encompass both Mode-I-dominant and mixed-mode geometries, broader statistical validation will require more datasets across different bridges and load environments. Furthermore, the current DT assumes deterministic material parameters; incorporating stochastic material variability and measurement noise will improve uncertainty quantification. Further automation and computational acceleration will be pursued to reduce the update turnaround time. Beyond the fracture-mechanics metrics used here, the updated DT can incorporate additional health indices such as strain ranges, traffic spectra, global vibration features, and deterioration indicators. Assimilating these data would reduce model uncertainty and enable more robust risk-based maintenance decisions, which we leave for future work. Future work should couple this physics-based confidence metric with Bayesian updating or data-assimilation frameworks, enabling real-time optimization of inspection frequency and maintenance strategies for long-span steel bridges.

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Code availability

The relevant code has been uploaded to GitHub (https://github.com/hollon-smith/Fatigue-crack-identification-and-quantification.git).

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Author contributions

X.L. and Z.F. conceived the research. X.L. and H.G. conducted the relevant design, modeling, and experimentation, and wrote the manuscript. B.J. and Z.X. provided relevant technical guidance for the research.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Zhongqiu Fu or Bohai Ji.

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